incur an aggregate estimate of 2,160 annual ongoing burden hours[1518] and $19,577,780 in annual ongoing external costs.[1519]

(ii) Comments/Responses on Ongoing Burden and Costs[1520]

No commenters suggested changes to the Commission's estimated ongoing burden hours and external costs that competing consolidators would incur in maintaining and operating their systems to comply with Rules 614(d)(1) through (d)(4). However, one commenter noted that the NYSE's ongoing costs associated with the NMS network are $215,000 per year,[1521] which is less than the burden hours and external costs the Commission preliminarily estimated a competing consolidator would incur for operating and maintaining a system to comply with Rules 614(d)(1) through (d)(4). As noted earlier, the Commission does not believe that the NMS network costs are directly applicable to the burden hour and cost estimates applicable to competing consolidators to build and operate systems to comply with Rules 614(d)(1) through (d)(4).

1518 The Commission estimated the monetized initial burden for this requirement to be $353,500. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Sr. Programmer at $332 for 210 hours) + (Sr. Systems Analyst at $285 for 180 hours) + (Compliance Manager at $310 for 60 hours) + (Director of Compliance at $489 for 30 hours) + (Compliance Attorney at $366 for 60 hours)] x [(4 SROs)] = 2,160 annual ongoing burden hours across the SROs and $705,000.

1519 The Commission estimated that the SROs would incur the following initial external costs: [($123,725 to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4)) + ($168,000 in monthly market data fees over 12 months) + ($4,602,720 to co-locate within four exchange data centers over 12 months)] x [(4 SROs)] = $19,577,780 across the SROs.

1520 One commenter stated that the costs to the industry may be significantly higher than the ongoing annual costs incurred by each competing consolidator because the proposal did not explain the fees the competing consolidators would charge investors. See Cboe Letter at 24.

1521 IDS Letter I at 13, n.38.

However, the Commission believes it is reasonable to increase its ongoing burden hour and external cost estimates to operate systems to collect, consolidate, and disseminate consolidated market data. As it did for its initial burden hour and external cost estimates, the Commission is increasing its estimated ongoing burden hours for Sr. Programmers and Sr. Systems Analysts by three times because competing consolidators may potentially build aggregation systems in three data centers, so they consequently must maintain these systems, as well as its estimated external costs associated with operating and maintaining systems by three times, for the same reason.

(iii)Adopted Estimates – Annual Ongoing and Costs

The Commission continues to believe that all types of competing consolidators would incur similar ongoing, annual burdens once their systems have been built because such systems would likely be similar in nature. As it did for its revised initial burden hour and external cost estimates, the Commission is increasing by three times its estimated ongoing burden hours for Sr. Programmers and Sr. Systems Analysts and external ongoing technology cost estimates because competing consolidators may potentially build aggregation systems in three data centers, and would have to maintain these systems. The Commission is also revising its total ongoing burden hour and external cost estimates to reflect a reduction in the number of potential broker-dealers that aggregate market data for internal uses and SRO competing consolidators.

The Commission estimates that each competing consolidator would incur 1,320 ongoing, annual burden hours[1522] and external costs of $371,175 to operate and maintain its systems to

---

[1522] The Commission estimates the monetized annual ongoing burden for this requirement to be $418,290. These estimates were based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 630 hours) + (Sr. Systems Analyst at $285/hour for 540 hours) + (Compliance Manager at $310/hour for

comply with Rules 614(d)(1)-(d)(4),[1523] as well as external ongoing, annual external costs of $4,602,720 for co-location connectivity to the exchange data centers,[1524] and $168,000 to purchase market data from the exchanges,[1525] for a total ongoing, annual external cost to each competing consolidator of $5,141,895.[1526]

The Commission estimates that the total ongoing, annual external burden hours to be incurred by market data aggregation firms, exclusive SIPs and new entrants would be 2,640

---

60 hours) + (Director of Compliance at $489/hour for 30 hours) + (Compliance Attorney at $366/hour for 60 hours)] = 1,320 ongoing, annual burden hours per competing consolidator to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1523 This estimate was originally based on discussions with a market participant and the Commission's understanding of hardware costs. The Commission has increased this estimated cost by three times because the Commission believes that competing consolidators would have to maintain aggregation systems in three data centers.

1524 This estimate was based on an estimated $95,890 in monthly co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers over 12 months.

1525 As it did in the Proposing Release, the Commission used the monthly market data access and redistribution fees currently charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000), multiplied by 12 months.

1526 The Commission estimates that each market data aggregation firm/broker-dealer that aggregates market data/exclusive SIP/new entrant/SRO would incur the following ongoing, annual external costs: [($371,175 to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4)) + ($168,000 to purchase market data) + ($4,602,720 for co-location connectivity within four exchange data centers)] = $5,141,895.

burden hours,[1527] for each of these categories of competing consolidator, as well as total ongoing, annual external costs of $10,283,790,[1528] for each of these categories of competing consolidator.

The Commission estimates that the total ongoing, annual external burden hours to be incurred by broker-dealers that aggregate market data and SROs would be 1,320 burden

---

1527 The Commission estimates the total monetized annual ongoing burden for this requirement to be $836,580. These estimates were based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 630 hours) + (Sr. Systems Analyst at $285/hour for 540 hours) + (Compliance Manager at $310/hour for 60 hours) + (Director of Compliance at $489/hour for 30 hours) + (Compliance Attorney at $366/hour for 60 hours)] x [(2 market data aggregation firms/exclusive SIPs/new entrants] = 2,640 total ongoing, annual burden hours to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4) for each of these categories of competing consolidator. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1528 The Commission estimates the total annual ongoing external cost for market data aggregation firms/exclusive SIPs/new entrants would be: [($371,175 to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4)) + ($168,000 to purchase market data) + ($4,602,720 for co-location connectivity within four exchange data centers)] x [(2 market data aggregation firms/exclusive SIPs/new entrants)] = $10,283,790 for each of these categories of competing consolidator.

hours,[1529] for each of these categories of competing consolidator, as well as total ongoing, annual external costs of $5,141,885,[1530] for each of these categories of competing consolidator.

4. Recordkeeping

Rule 614(d)(7) requires each competing consolidator to keep and preserve at least one copy of all documents made or received by it in the course of its business and in the conduct of its business. These documents must be kept for a period of no less than five years, the first two years in an easily accessible place. Rule 614(d)(8) requires each competing consolidator to furnish promptly these documents to any representative of the Commission upon request.

1529 The Commission estimates the total monetized annual ongoing burden for this requirement to be $418,290. These estimates were based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 630 hours) + (Sr. Systems Analyst at $285/hour for 540 hours) + (Compliance Manager at $310/hour for 60 hours) + (Director of Compliance at $489/hour for 30 hours) + (Compliance Attorney at $366/hour for 60 hours)] x [(1 broker-dealer that aggregates market data/SRO)] = 1,320 total ongoing, annual burden hours to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4) for each of these categories of competing consolidator. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1530 The Commission estimates the total annual ongoing external cost for broker-dealers that aggregate market data/SROs would be: [($371,175 to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4)) + ($168,000 to purchase market data) + ($4,602,720 for co-location connectivity within four exchange data centers)] x [(1 broker-dealer that aggregates market data/SRO)] = $5,141,885 for each of these categories of competing consolidator.

(a) Initial Burden and Costs

(i) Proposed Estimates – Initial Burden and Costs

In the Proposing Release, the Commission estimated that these requirements would create an initial burden of 40 hours (for a total cost of $8,720),[1531] for a total initial burden of 480 hours for all respondents (for a total cost of $104,640). These estimates were based on the Commission's experience with recordkeeping costs and consistent with prior burden estimates for similar provisions.[1532]

(ii) Comments/Responses on Initial Burden and Costs

The Commission did not receive any comments on the estimated initial burdens and costs of Rule 614(d)(7) and Rule 614(d)(8).

(iii)Adopted Estimates – Initial Burden and Costs

The Commission is revising its preliminary estimates to account for the downward estimate from 12 competing consolidators to 8 competing consolidators. The Commission estimates that the initial burden of 40 hours (for a total cost of $8,720) for a total initial burden of 320 hours for all respondents (for a total cost of $69,760) is reasonable based upon the Commission's experiences with estimating similar provisions.

1531 The Commission based this estimate on the $218 hourly rate as of May 2019 for a paralegal x 40 hours. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1532 See Security Based Swap Data Repository Registration, Duties, and Core Principles, Securities Exchange Act Release No. 74246 (Feb. 11, 2015), 80 FR 14438 (Mar. 19, 2015) at 14541.

(b) Ongoing Burden and Costs

(i) Proposed Estimates – Ongoing Burden and Costs

The Commission estimated that the ongoing annual burden of recordkeeping in accordance with Rules 614(d)(7) and 614(d)(8) would be 20 hours per respondent (for a total cost of $4,360) and a total ongoing annual burden of 240 hours for all respondents (for a total cost of $52,320).

(ii) Comments/Responses on Ongoing Burden and Costs

The Commission did not receive any comments on the estimated ongoing burdens and costs of Rule 614(d)(7) and Rule 614(d)(8).

(iii)Adopted Estimates – Ongoing Burden and Costs

The Commission is revising its preliminary estimates to account for the downward estimate from 12 competing consolidators to eight competing consolidators. The Commission estimates that the ongoing burden of 20 hours (for a total cost of $4,360) for a total initial burden of 160 hours for all respondents (for a total cost of $34,880) is reasonable based upon the Commission's experiences with estimating similar provisions.

5. Reports and Reviews

Rule 614(d)(5) and (d)(6) requires competing consolidators to produce monthly reports on performance metrics and systems issues.

(a) Initial Burden and Costs

(i) Proposed Estimates – Initial Burden and Costs

The Commission estimated that the average one-time, initial burden to program systems to produce the monthly reports required by Rules 614(d)(5) and (d)(6), including keeping the information publicly posted and free and accessible (in downloadable files under Rule

614(d)(5)), would be 246 hours per competing consolidator (for a total cost of $80,507)[1533] and $800 in external costs.[1534] The Commission estimated that the total initial burden would be 2,952 hours (for a total cost of $966,804)[1535] and a total initial external cost of $9,600.[1536]

1533 This figure was based on the estimated initial paperwork burden for 17 CFR 242.606(a) (Rule 606(a)), which requires each broker or dealer to make publicly available on a website a quarterly report on its routing of non-directed orders in NMS stocks that are submitted on a held basis and of non-directed orders that are customer orders in NMS securities. See Securities Exchange Act Release No. 84528 (Nov. 2, 2018), 83 FR 58338 (Nov. 19, 2018) ("Order Handling Disclosure Release"). In the Proposing Release, the Commission converted the 10 hour estimate for a quarterly report to an estimate for a monthly report. In addition, the Commission added the burden of posting the required information to the website. The Commission estimated the monetized initial burden for this requirement to be $80,507. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Sr. Programmer at $332 per hour for 160 hours) + (Sr. Database Administrator at $342 per hour for 20 hours) + (Sr. Business Analyst at $275 per hour for 20 hours) + (Attorney at $417 per hour for 4 hours) + (Sr. Operations Manager at $366 per hour for 20 hours) + (Systems Analyst at $263 per hour for 16 hours) + ($308.50 blended rate for Sr. Systems Analyst and Sr. Programmer for 6 hours)] = 246 initial burden hours per competing consolidator and $80,507.

1534 The Commission estimated that each competing consolidator would incur an initial external cost of $800 for an external website developer to create the website.

1535 The Commission estimated the monetized initial aggregate burden for this requirement to be $966,804. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Sr. Programmer at $332 per hour for 160 hours) + (Sr. Database Administrator at $342 per hour for 20 hours) + (Sr. Business Analyst at $275 per hour for 20 hours) + (Attorney at $417 per hour for 4 hours) + (Sr. Operations Manager at $366 per hour for 20 hours) + (Systems Analyst at $263 per hour for 16 hours) + ($308.50 blended rate for Sr. Systems Analyst and Sr. Programmer for 6 hours)] x [(12 competing consolidators)] = 2,952 initial aggregate burden hours across the competing consolidators and $966,804.

1536 $9,600 = ($800 for an external website developer to create the website) x (12 competing consolidators).

(ii) Comments/Responses on Initial Burden and Costs

The Commission did not receive any comments on the estimated initial burdens and costs of Rule 614(d)(5) and Rule 614(d)(6).

(iii)Adopted Estimates – Initial Burden and Costs

The Commission is revising its preliminary estimates to account for the downward estimate from 12 competing consolidators to eight competing consolidators. The Commission estimates that the initial burden of 246 hours per competing consolidator (for a total cost of $80,507)[1537] and $800 in external costs.[1538] The Commission estimates that the total initial

---

1537 This figure was based on the estimated initial paperwork burden for Rule 606(a), which requires each broker or dealer to make publicly available on a website a quarterly report on its routing of non-directed orders in NMS stocks that are submitted on a held basis and of non-directed orders that are customer orders in NMS securities. See Order Handling Disclosure Release, supra note 1533. In the Proposing Release, the Commission converted the 10 hour estimate for a quarterly report to an estimate for a monthly report. In addition, the Commission added the burden of posting the required information to the website. The Commission estimated the monetized initial burden for this requirement to be $80,507. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Sr. Programmer at $332 per hour for 160 hours) + (Sr. Database Administrator at $342 per hour for 20 hours) + (Sr. Business Analyst at $275 per hour for 20 hours) + (Attorney at $417 per hour for 4 hours) + (Sr. Operations Manager at $366 per hour for 20 hours) + (Systems Analyst at $263 per hour for 16 hours) + ($308.50 blended rate for Sr. Systems Analyst and Sr. Programmer for 6 hours)] = 246 initial burden hours per competing consolidator and $80,507.

1538 The Commission estimated that each competing consolidator would incur an initial external cost of $800 for an external website developer to create the website.

burden would be 1,968 hours (for a total cost of $644,056)[1539] and a total initial external cost of $6,400.[1540]

(b) Ongoing Burden and Costs

(i) Proposed Estimates – Ongoing Burden and Costs

The Commission estimated that each competing consolidator would incur an average burden of 11 hours to prepare and make publicly available a monthly report in the format required by Rules 614(d)(5) and (d)(6) (for a total cost of $3,768.50), or a burden of 132 hours per year (for a total cost of $45,222).[1541] Once a report is posted on an internet website, the

---

1539 The Commission estimates the monetized initial aggregate burden for this requirement to be $644,056. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Sr. Programmer at $332 per hour for 160 hours) + (Sr. Database Administrator at $342 per hour for 20 hours) + (Sr. Business Analyst at $275 per hour for 20 hours) + (Attorney at $417 per hour for 4 hours) + (Sr. Operations Manager at $366 per hour for 20 hours) + (Systems Analyst at $263 per hour for 16 hours) + ($308.50 blended rate for Sr. Systems Analyst and Sr. Programmer for 6 hours)] x [(8 competing consolidators)] = 1,968 initial aggregate burden hours across the competing consolidators and $644,056.

1540 $6,400 = ($800 for an external website developer to create the website) x (8 competing consolidators).

1541 This figure was based on the estimated ongoing paperwork burden for Rule 606(a), which requires each broker or dealer to make publicly available on a website a report on a quarterly basis. In the Paperwork Reduction Act discussion for Rule 606(a), the Commission established that the average annual burden for a broker-dealer to comply with Rules 606(a)(1)(i) through (iii) would be 10 hours. See Order Handling Disclosure Release, supra note 1533. In the Proposing Release, the Commission converted the 10 hour estimate for a quarterly report to an estimate for a monthly report. In addition, the Commission added the burden of updating the website. The Commission estimated the monetized annual burden for this requirement to be $3,768.50. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Sr. Business Analyst at $275 per hour for 5